



For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY
PAPER 3

6031/3
2 hours 30 minutes

JUNE 2025 SESSION

Additional materials:
Answer paper
Data Booklet
Electronic calculator
Graph paper

INSTRUCTIONS TO CANDIDATES

Answer six questions.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.



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This question paper consists of 12 printed pages.

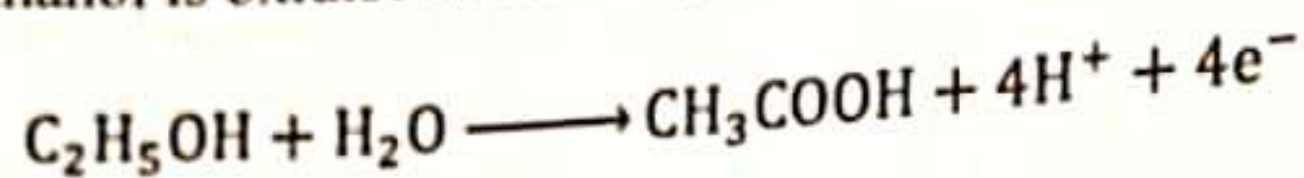
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Section A

Answer any two questions from this section.

- 1 (a) A volume of 1.00 cm^3 of a hand sanitiser was diluted to 25.00 cm^3 using $1.00 \text{ mol dm}^{-3} \text{ H}_2\text{SO}_4$. The resulting mixture required 28.00 cm^3 of 0.40 mol dm^{-3} of acidified KMnO_4 for titration.

- (i) Identify, with a reason, a suitable indicator for the titration.
- (ii) Ethanol is oxidised according to the equation:



Write a balanced chemical equation for the reaction that occurs during the titration.

- (iii) Calculate the number of moles of ethanol in

1. 1.00 cm^3 of the hand sanitiser.
2. $1\,000 \text{ cm}^3$ of 100% ethanol,
3. $1\,000 \text{ cm}^3$ of the hand sanitiser,

- (iv) Hence calculate the percentage by volume of ethanol in $1\,000 \text{ cm}^3$ of the hand sanitiser.

[The density of 100% ethanol is 925 g/dm^3 .]

[10]

- (b) A volume of 10.00 cm^3 of a gaseous hydrocarbon was mixed with 300 cm^3 of air and exploded. The volume decreased to 280 cm^3 . The remaining gas was bubbled through concentrated $\text{KOH}_{(\text{aq})}$ and the volume decreased further to 250 cm^3 .

Calculate the

- (i) volume of $\text{CO}_{2(\text{g})}$ in the gas mixture after explosion,
- (ii) empirical formulae of the hydrocarbon.

[5]

[Total:15]



- (a) A total of 13.65 kJ of energy were evolved when 2.15 g of magnesium were reacted with excess nitrogen.

- (i) Write a balanced chemical equation for the reaction.
- (ii) Calculate the
 1. amount of magnesium nitride formed,
 2. enthalpy change of formation of magnesium nitride.

[3]

- (b) Table 2.1 gives enthalpy information of the compounds shown.

Table 2.1

	LiCl	LiF
lattice enthalpy/kJmol ⁻¹	-843	-1029
sum of hydration enthalpies/kJmol ⁻¹	-883	-1023

- (i) Calculate the enthalpy change of solution of each of the compounds.
- (ii) Comment on the solubilities of each of the compounds in water.

[3]

- (c) (i) Name the type of catalysis used in the contact process.
- (ii) Describe the mode of catalysis.

[3]

- (d) Explain why

- (i) carbon graphite is used as a lubricant,
- (ii) the bond angles of Group (V) hydrides are larger than those of the corresponding Group (VI) hydrides,
- (iii) solid water floats on liquid water.

[6]

[Total:15]



- 3 (a) The mass spectrum of magnesium ions gives peaks at m/e 24, 25 and 26. The respective percentage abundances are 78.99, 10 and 11.01.

- Draw a sketch graph of the spectrum of magnesium.
- Suggest the reason for the appearance of only 3 peaks in the spectrum.
- Calculate the relative atomic mass of the magnesium element.

[6]

- (b) Explain why the first ionisation energy of magnesium is

- higher than that of sodium,
- lower than its third ionisation energy.

[2]

- (c) Calculate the lattice enthalpy of aluminium oxide given that:

$$\Delta H_f^\theta(\text{Al}_2\text{O}_3) = -1\,669.8 \text{ kJ mol}^{-1}$$

$$\Delta H_{at}^\theta(\text{Al}) = +32.6 \text{ kJ mol}^{-1}$$

$$\Delta H_{EA}^\theta(\text{O}) = -1\,42 \text{ kJ mol}^{-1}$$

$$\Delta H_{EA}^\theta(\text{O}^-) = +844 \text{ kJ mol}^{-1}$$

[use of relevant data from the data booklet is recommended]

[7]

[Total:15]



Section B

Answer *one* question from this section.

- 4 (a) Explain why
- (i) solubility of Group (II) oxides increases down the group,
 - (ii) thermal stability of Group (II) carbonates increases down the group.
- [3]
- (b) Describe and explain the differences in the reactions of concentrated sulphuric acid with
- (i) NaCl,
 - (ii) NaBr,
 - (iii) NaI.
- [6]
- (c) Write balanced chemical equations to show what happens when
- (i) astatine is added to aqueous sodium hydroxide at room temperature,
 - (ii) aqueous silver nitrate is added to aqueous sodium astatine.
- [4]
- (d) Table 4.1 shows the pKa values of Group (VII) halides.

Table 4.1

	HF	HCl	HBr	HI
pKa	3.25	-7.4	-9.5	-10

Explain the trend in the pKa values. [2]

[Total:15]



- 5 (a) Table 5.1 shows the melting points of Group (IV) oxides.

Table 5.1

oxide	CO ₂	SiO ₂	GeO ₂	SnO ₂	PbO ₂
melting point/°C	-56	1 610	1 116	1127	decomposed at 300°

- (i) Explain the acid-base nature of each of the oxides. [10]
- (ii) Comment on the differences in the melting points of the oxides.
- (b) (i) State the conditions for the Haber process.
- (ii) Explain how increasing temperature, during the Haber process, affects the
- equilibrium position of the reaction,
 - rate of the reaction.
- [Total: 15]



Section C

Answer any two questions from this section.

- 6 (a) Fig 6.1 shows some of the compounds, F and G, present in a liquid fuel.

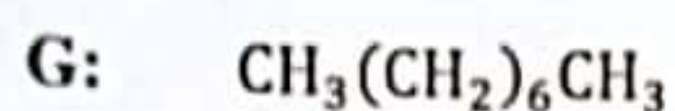
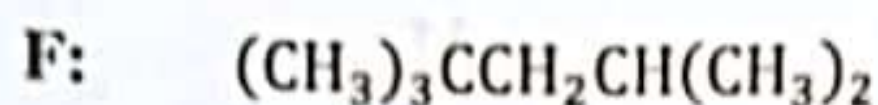


Fig 6.1

- (i) Give the IUPAC name of
1. compound F,
 2. compound G.
- (ii) Describe and explain how a mixture of F and G can be separated in a laboratory.
- (iii) Explain why F is suitable for use as a fuel additive. [5]
- (b) Explain each of the following observations:
- (i) ethanamide, CH_3CONH_2 is a neutral molecule
 - (ii) ethanoyl chloride, CH_3COCl is more acidic in water than ethanoic acid
 - (iii) the solution formed when 2-chloropropanal $\text{CH}_3\text{CH}(\text{Cl})\text{CHO}$ is hydrolysed is optically inactive
 - (iv) $\text{CH}_2\text{Cl}_2\text{CHF}_2$ is non flammable
 - (v) tetrachloroethene, C_2Cl_4 is a suitable dry-cleaning solvent [10]



7

Structures of compounds **K**, **M** and **N** are shown in Fig. 7.1

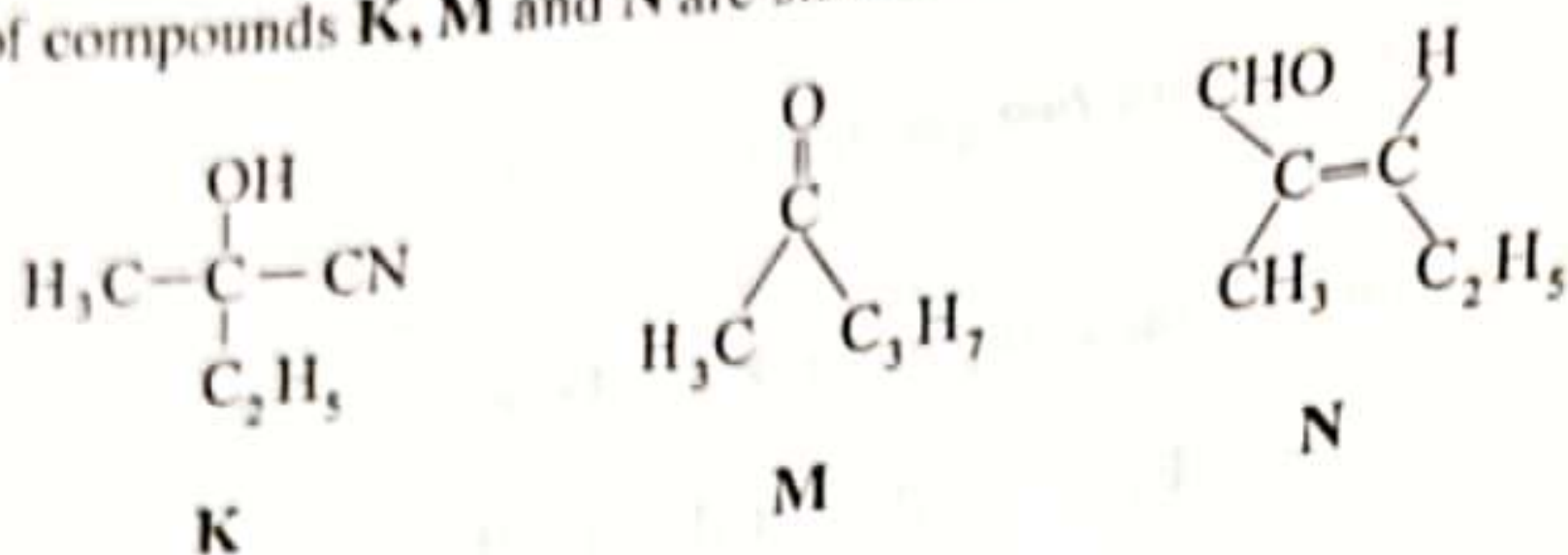


Fig 7.1

- (a) Give the structures of the organic product formed when
- (i) **K** reacts with acidified dichromate,
 - (ii) **M** reacts with 2,4-dinitrophenylhydrazine,
 - (iii) **N** reacts with dilute alkaline KMnO_4 . [3]
- (b) The reaction scheme shown in Fig 7.2 shows how new functional groups can be added to organic compounds.

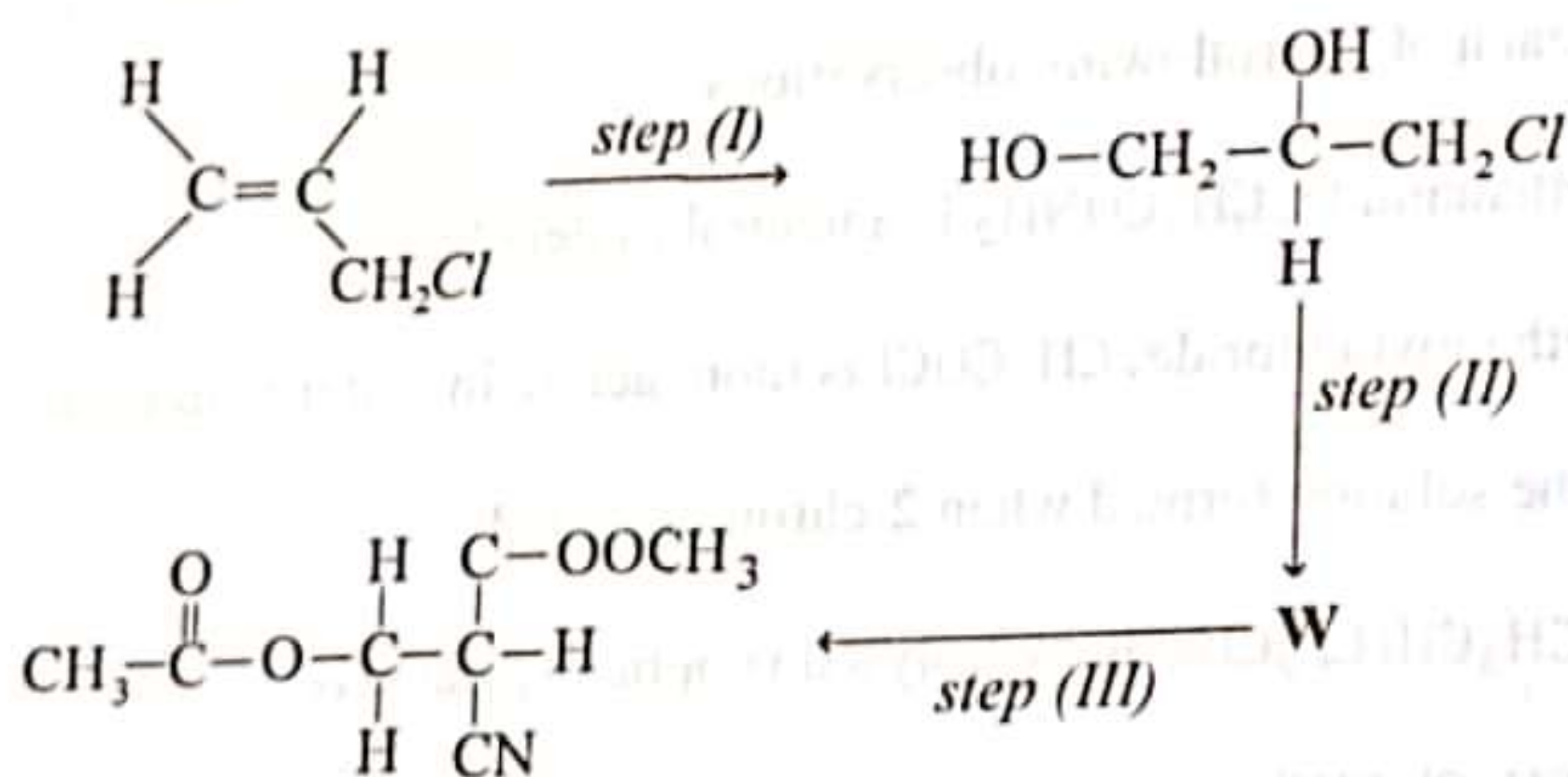


Fig 7.2

- (i) Suggest reagents and conditions for *steps I, II and III*.
 - (ii) Deduce the structural formula of **W**. [7]
- (c) Explain why
- (i) beer turns sour when left to stand in air,
 - (ii) the smell of fish disappears when fresh lemon juice is added. [5]

[Total: 15]

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- 8 (a) Fig 8.1 shows how compounds **B** and **C** can be formed under suitable conditions. **B** gives a positive iodoform test and **C** reacts with aqueous bromine.

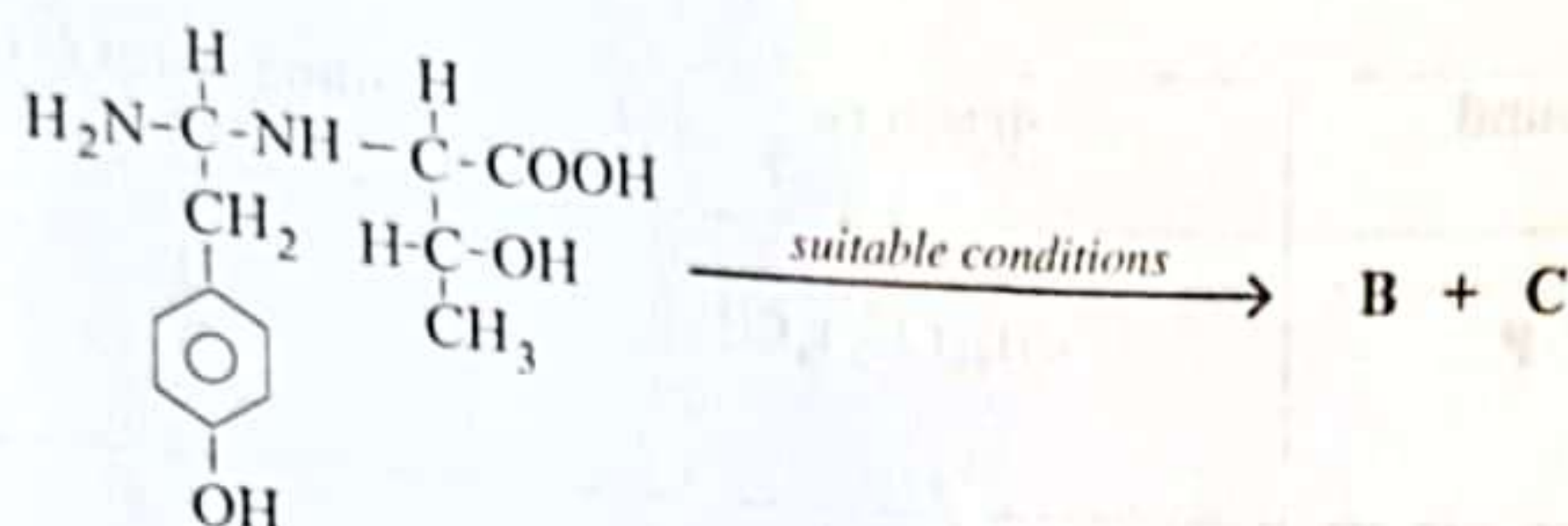


Fig. 8.1

- (i) Suggest the suitable conditions for the formation of **B** and **C**.
- (ii) Draw the possible structure of
 1. **B**,
 2. **C**.
- (iii) Name the class of organic compounds to which **B** and **C** belongs.
- (iv) Suggest, with reasons, a method of separating a mixture of **B** and **C**.
- (v) State the observable changes that occur when **C** reacts with aqueous bromine.

[9]



- (b) Table 8.1 shows boiling points of three organic compounds, P, Q and R.

Table 8.1

compound	structure	boiling point / °C
P	$\text{CH}_3(\text{CH}_2)_3\text{CH}_3$	31
Q	$ \begin{array}{c} \text{CH}_3 \quad \text{CH}_3 \\ \diagdown \quad \diagup \\ \text{C} = \text{C} \\ \diagup \quad \diagdown \\ \text{CH}_3 \quad \text{H} \end{array} $	22
R	$ \begin{array}{c} \text{CH}_3 \quad \text{CH}_3 \\ \diagdown \quad \diagup \\ \text{C} \\ \diagup \quad \diagdown \\ \text{CH}_3 \quad \text{CH}_3 \end{array} $	11

- (i) Explain the differences in the boiling points of P, Q and R.
- (ii) Fig 8.2 shows how compound Q is formed from an organic compound S.

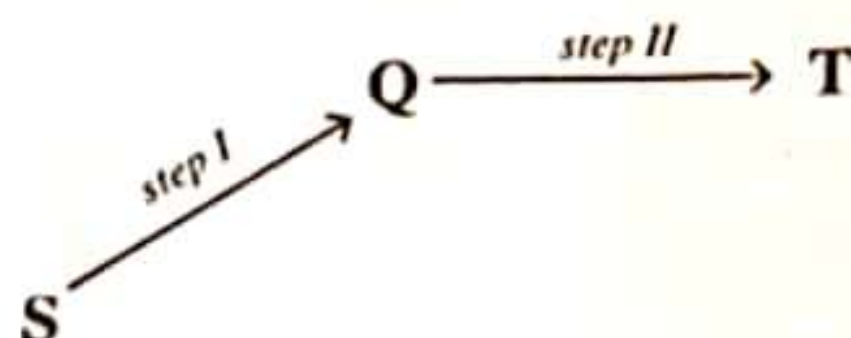


Fig 8.2

State the reagents and conditions for *steps I* and *II*, given that S is a halogenoalkane and T is a diol.

- (iii) Draw the structure of the organic product formed when Q reacts with steam.

[6]

[Total:15]

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Section D

Answer **one** question from this section.

- 9 (a) (i) State any **two** disadvantages of land filling.
- (ii) Identify, with reasons, any **two** suitable criteria for selecting landfill sites.
- (iii) Describe how invasion by pests and small animals is minimised in landfill sites.
- (b) (i) Explain why electrophoresis is pH dependent.
- (ii) Fig 9.1 shows structures of three amino acids A, B and C.

[5]

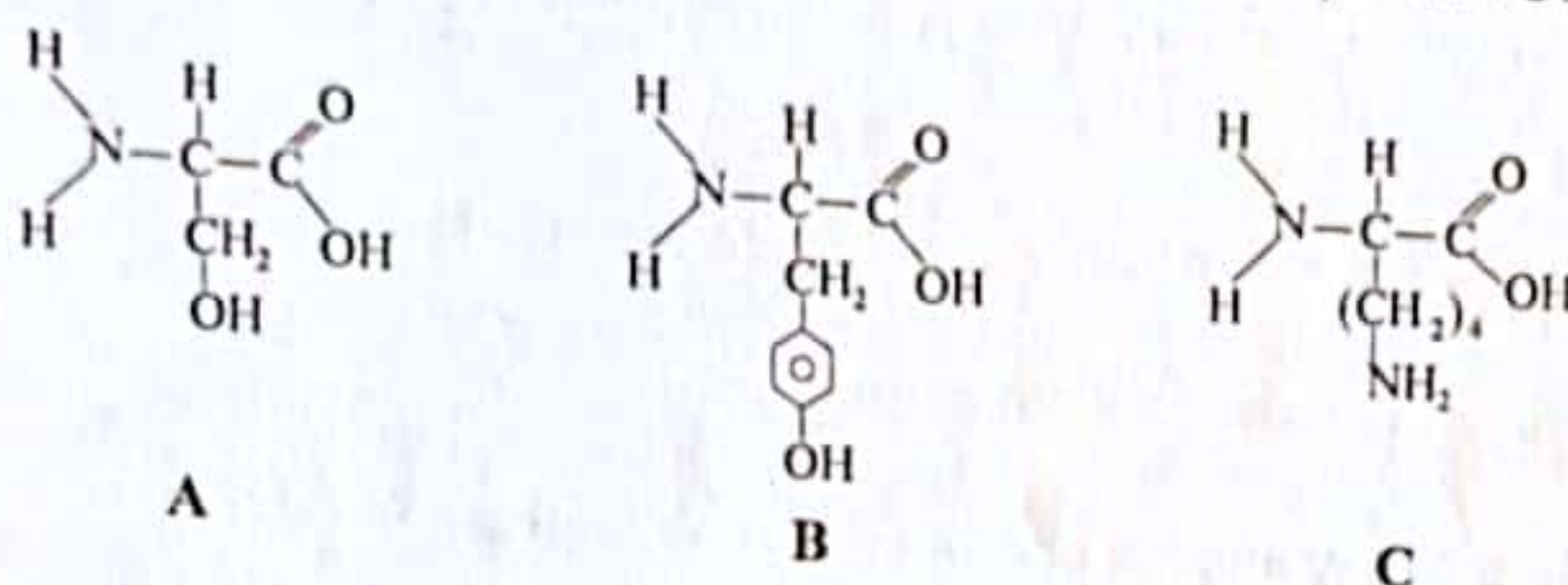


Fig. 9

Compare the relative migration of the amino acids at pH 4 and at pH 12.

[6]

- (c) Show the isomeric forms of
- (i) diamine dichloroplatinum,
- (ii) *tris*-ethanedioate chromium (III).

[4]

[Total: 15]



- 10 (a) (i) State the advantage, in solvent extraction, of using successive extractions with smaller volumes of solvent than one large volume.
- (ii) Describe how a separating funnel is used in solvent extraction.
- (iii) Give one application of solvent extraction. [5]

(b) The $\text{Fe}^{2+}/\text{MnO}_4^-$ titration is used in chemical analysis.

- (i) Name the ion responsible for the colour at end point.
- (ii) Suggest an identity of the brown suspension that is formed in the titration flask when left to settle.
- (iii) The compound in (ii) is used as a depolariser in dry cells. Write an equation for the depolarisation process. [3]

- (c) (i) Explain how separation is achieved in chromatography.
- (ii) Name the stationary phase used in GLC.
- (iii) Give two examples of a suitable mobile phase in GLC.
- (iv) Explain how components in GLC separation are

1. identified,
2. quantified.

[7]

[Total:15]



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